

Project Design Phase II: Data Flow Diagram & User Stories

Automated Network Request Management in ServiceNow

Date	30 July 2026	Team ID	Individual / Solo Project
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Data Flow Diagrams (DFD)

A Data Flow Diagram is a traditional visual representation of information flows within a system. It shows how data enters and leaves the system, what processes transform the information, and where data is stored.

For the **Automated Network Request Management** project, a DFD helps clarify the flow of request data, approval decisions, security policy validation, and deployment commands across ServiceNow, the CMDB, and network automation systems.

System Overview

The system receives network access requests from employees via the Service Catalog, routes them through Flow Designer for approval decisions (checking with the CMDB for valid Configuration Items and active policies), updates topology relationships, and triggers deployment through IntegrationHub to network automation tools like Ansible. Below is a description of the major data flows and processes for the Level 0 DFD:

- **Key Entities:** The Employee (actor), Service Portal (interface), Flow Designer (decision logic), CMDB (topology & CI validation), IntegrationHub (deployment orchestration), and Ansible/Network Controllers (infrastructure management).
- **Data Flows:** Original network request parameters (source/destination IPs, ports), manager/InfoSec approval decisions, CMDB validation checks, deployment commands to network devices, and confirmation callbacks to close the request.

User Stories

User stories capture the functional requirements from the perspective of each stakeholder role. They follow the template: *"As a [role], I want to [action] so that [benefit]."* Below are the key user stories for the Automated Network Request Management system:

User Type	Epic	US ID	User Story	Priority	Release
Employee	Catalog Discovery	EMP-1	As an employee, I want to find the Network Request Catalog item on the Service Portal so I can request firewall or port changes I need.	High	MVP
Employee	Catalog Request	EMP-2	As an employee, I want to fill out a request form (source IP, destination IP, port, business justification) and submit it so I can formally request network access.	High	MVP
Employee	Status Tracking	EMP-3	As an employee, I want to see the real-time status of my network request (pending, approved, deploying, completed) so I know where my request stands.	High	MVP
Employee	Notifications	EMP-4	As an employee, I want to receive email and/or portal notifications when my request is approved, rejected, or deployed so I don't have to keep checking manually.	High	MVP
Manager	Approval	MGR-1	As a manager, I want to receive approval requests in my worklist so I can review and approve/reject network access requests from my team.	High	MVP
Manager	Approval Context	MGR-2	As a manager, I want to see the employee name, requested network change details, and business justification on the approval request so I can make an informed decision.	High	MVP
Manager	Escalation	MGR-3	As a manager, if I do not approve/reject within 3 days, the request should escalate to my skip-level manager so critical network requests aren't blocked.	Medium	Phase-2
Network Admin	Fulfillment	IT-1	As a Network Admin, I want approved requests to automatically trigger a fulfillment task and deploy the configuration via IntegrationHub/Ansible so I don't have to manually configure devices.	High	MVP

User Type	Epic	US ID	User Story	Priority	Release
Network Admin	CMDB Integration	IT-2	As a Network Admin, I want the system to check the CMDB for valid CIs and automatically update topology relationships when an approval happens so our inventory stays accurate.	High	Phase-2
Network Admin	Failure Handling	IT-3	As a Network Admin, if a configuration deployment fails, I want to be notified immediately with error details so I can troubleshoot and retry.	High	MVP

These user stories form the foundation of the design and development phases. They ensure that the solution addresses the needs of each stakeholder and delivers value across the employee request experience, manager approval workflow, and network fulfillment automation.